



United International University

Department of Computer Science & Engineering

Mid-Term Exam, Trimester: Summer 2025

Course Code: CSE 3715

Course Title: Data Communication

Duration: 1 hour 30 Minutes

Total Marks: 30

(Any examinee found adopting unfair means would be expelled from the trimester/program as per UIU disciplinary rules.)

Answer all the questions

1. A university plans a data communication system where the top priority is conducting live online classes, and the second priority is instant messaging between teachers and students.

(a) **Identify** and **explain** the characteristics that determine the effectiveness of data communication for this network. [3]
(CO1)

(b) **Justify** whether a Bus topology would be suitable or unsuitable for this system. [2]
(CO1)

2. The Open System Interconnection (OSI) model uses different types of addresses to enable reliable communication:

- to uniquely identify devices within a local network,
- to locate devices across a larger network, and
- to distinguish between different applications or processes on a device.

Data is transmitted reliably using error control and flow control at different layers, and additional information is added at each stage to ensure it reaches the correct destination.

(a) **Explain** why error control and flow control are implemented at two separate layers of the communication system. [2]
(CO1)

(b) **Describe** how encapsulation works in this communication, including the addition of headers and trailers. [3]
(CO1)

3. A composite signal is composed of the following four waves:

(i) $x_1(t) = -2 \sin(50\pi t + 40)$

(iii) $x_3(t) = -8 \sin(400\pi t + 42)$

(ii) $x_2(t) = 4 \sin(100\pi t + 41)$

(iv) $x_4(t) = 16 \sin(200\pi t + 180)$

The signal power is 31 mW and the noise power is 11 μ W. It passes through a noisy channel which has a bandwidth of 4.17 kHz.

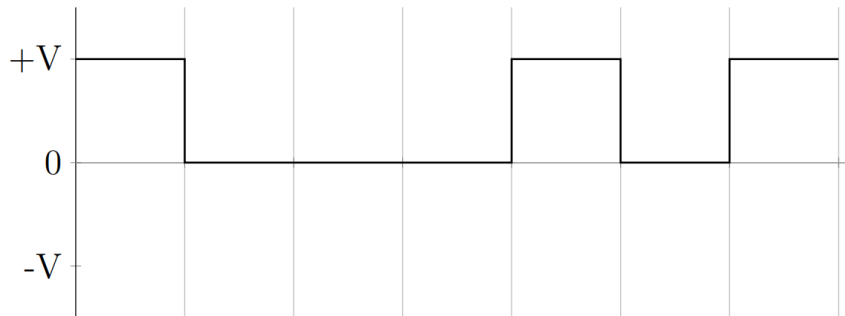
(a) **Calculate** the bandwidth of the composite signal and **demonstrate** the spectrum of $x(t)$ using frequency domain. [2]
(CO2)

(b) **Calculate** the channel capacity and **determine** the minimum number of signal levels required for transmission if the system utilizes only 70% of the channel capacity. [5]
(CO2)

4. A digital signal is initially transmitted using bipolar AMI but found to be faulty. It was later re-transmitted using Manchester line coding.

(a) **Examine** the AMI signal shown in the figure and **explain** why it was considered faulty.

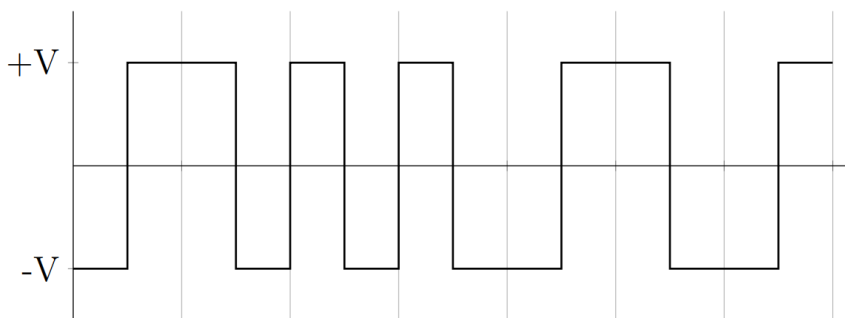
[2]
(CO3)



(b) The following signal is transmitted using Manchester coding. **Determine** the original data sequence and **illustrate** how the same data would appear if transmitted using Polar RZ and NRZ-I line coding schemes.

(Assume the last signal level was positive.)

[5]
(CO3)



(c) **Explain** why multiple line coding schemes are used and what problems they address.

[3]
(CO3)

5. **Convert** the last two digits of your student ID into a eight-bit binary number.

Then, **illustrate** with a diagram how this data would be modulated using Binary Amplitude Shift Keying (ASK) and Binary Phase Shift Keying (PSK).

[3]
(CO3)